

Experiment 6 — The Sleuth Kit (TSK)

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Aim:

To use The Sleuth Kit (TSK), a collection of command-line tools, to analyze a disk image (4Dell Latitude CPi.E01), examine its file system, and recover digital evidence.

Prerequisites and Installation:

1. Sleuth Kit Tools: Download the Windows version of Sleuth Kit (sleuthkit-4.14.0-win32.zip) and extract it to a known location, for example, C:\SleuthKit.

C:\Users\sleuthkit-4.14.0-win32 2. Evidence Files: Download the sample disk image files (4Dell Latitude CPi.E01 and 4Dell Latitude CPi.E02) from the Google Drive link provided in the lab manual.

3. Case Folder: Create a dedicated folder for your investigation, for example:

C:\Forensics_Lab

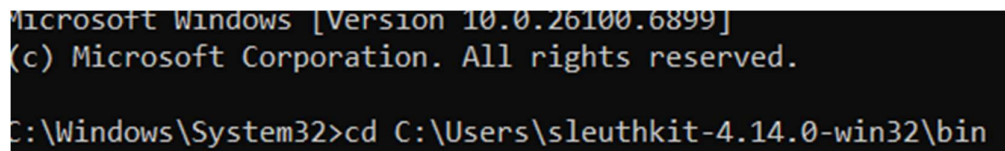
Place the evidence files inside this folder.

Procedure:

Step 1: Open Command Prompt and Navigate

- Open Command Prompt (cmd.exe) as Administrator.
- Navigate to the bin directory inside your Sleuth Kit folder. This directory contains all the executable tools (.exe).

Example command: C:\Windows\System32> cd C:\Users\sleuthkit-4.14.0-win32\bin



```
Microsoft Windows [Version 10.0.26100.6899]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd C:\Users\sleuthkit-4.14.0-win32\bin
```

Step 2: List Partitions (mmls)

- Use the mmls (list partitions) command to view the partition table of the disk image. This helps identify the offset (starting sector) of the partition to be analyzed.
- The path to the image file must be enclosed in quotes.

Example command: C:\Users\sleuthkit-4.14.0-win32\bin> mmls.exe "C:\Forensics_Lab\4Dell Latitude CPi.E01"

Tip: The output displays the partition details. The NTFS or exFAT (0x07) partition (Slot 002) usually starts at sector 63. This offset value will be required in the next steps.

```

Microsoft Windows [Version 10.0.26100.6899]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd C:\Users\sleuthkit-4.14.0-win32\bin

C:\Users\sleuthkit-4.14.0-win32\bin>mmls.exe "C:\Forensics_Lab\4Dell Latitude CPi.E01"
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

```

	Slot	Start	End	Length	Description
000:	Meta	0000000000	0000000000	0000000001	Primary Table (#0)
001:	-----	0000000000	0000000062	0000000063	Unallocated
002:	000:000	0000000063	0009510479	0009510417	NTFS / exFAT (0x07)
003:	-----	0009510480	0009514259	0000003780	Unallocated

Step 3: Analyze File System (fsstat)

- Use the fsstat (file system statistics) command to get detailed information about the partition.
- Use the -o 63 flag to specify the partition offset.
- The command displays the result on-screen.

```
C:\Users\sleuthkit-4.14.0-win32\bin> fsstat.exe -o 63 "C:\Forensics_Lab\4Dell Latitude CPi.E01"
```

```

C:\Users\sleuthkit-4.14.0-win32\bin>fsstat.exe -o 63 "C:\Forensics_Lab\4Dell Latitude CPi.E01"
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: B26CB1CE6CB18D9B
DEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 2097152
First Cluster of MFT Mirror: 4755208
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 12305
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 9510415
Total Sector Range: 0 - 9510415

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16)  Size: 48-72  Flags: Resident
$ATTRIBUTE_LIST (32)       Size: No Limit  Flags: Non-resident
$FILE_NAME (48)            Size: 68-578   Flags: Resident,Index
$OBJECT_ID (64)            Size: 0-256    Flags: Resident
$SECURITY_DESCRIPTOR (80)   Size: No Limit  Flags: Non-resident
$VOLUME_NAME (96)          Size: 2-256   Flags: Resident
$VOLUME_INFORMATION (112)   Size: 12-12   Flags: Resident
$DATA (128)                Size: No Limit  Flags:
$INDEX_ROOT (144)          Size: No Limit  Flags: Resident
$INDEX_ALLOCATION (160)     Size: No Limit  Flags: Non-resident
$BITMAP (176)              Size: No Limit  Flags: Non-resident
$REPARSE_POINT (192)       Size: 0-16384   Flags: Non-resident
$EA_INFORMATION (208)      Size: 8-8     Flags: Resident
$EA (224)                  Size: 0-65536   Flags:
$LOGGED_UTILITY_STREAM (256) Size: 0-65536   Flags: Non-resident

C:\Users\sleuthkit-4.14.0-win32\bin>_

```

Step 4: List Files and Directories (fls)

- Use the fls (file list) command to recursively list all files and directories in the partition.
- Use the -r flag for recursion and redirect (>) the output to a text file for easier review.
- No terminal output will appear after execution.

```
C:\Users\sleuthkit-4.14.0-win32\bin> fls.exe -r -o 63 "C:\Forensics_Lab\4Dell Latitude CPi.E01" > C:\Forensics_Lab\file_list.txt
```

 **Tip:** You can open the file_list.txt file in your case folder to view the complete directory structure.

```
C:\Users\sleuthkit-4.14.0-win32\bin> fls.exe -r -o 63 "C:\Forensics_Lab\4Dell Latitude CPi.E01" > C:\Forensics_Lab\file_list.txt
```

Step 5: Analyze File Metadata (istat)

- Use the istat (inode statistics) command to display metadata about a specific file, such as MAC times (Modified, Accessed, Changed).
- From file_list.txt, identify an interesting file — here, Mr. Evil.bmp.
- Its inode number is 9871.
- Redirect the output to a metadata report file.

```
C:\Users\sleuthkit-4.14.0-win32\bin> istat.exe -o 63 "C:\Forensics_Lab\4Dell Latitude CPi.E01" 9871 > C:\Forensics_Lab\metadata_Mr_Evil.txt
```

```
C:\Users\sleuthkit-4.14.0-win32\bin> istat.exe -o 63 "C:\Forensics_Lab\4Dell Latitude CPi.E01" [inode_number] > C:\Forensics_Lab\metadata_info.txt
Invalid inode number: [inode_number] usage: istat.exe [-N num] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] [-P pooltype] [-B pool_volume_block] [-z zone] [-s seconds] [-rV] image inum
-N num: Force the display of MM address of block pointers
-r: display run list instead of list of block addresses
-z zone: time zone of original machine (i.e. EST/EDT or GMT)
-s seconds: Time skew of original machine (in seconds)
-i imgtype: The format of the image file (use '-i list' for supported types)
-b dev_sector_size: The size (in bytes) of the device sectors
-f fstype: File system type (use '-f list' for supported types)
-o imgoffset: The offset of the file system in the image (in sectors)
-P pooltype: Pool container type (use '-P list' for supported types)
-B pool_volume_block: Starting block (for pool volumes only)
-S snap_id: Snapshot ID (for APFS only)
-v: verbose output to stderr
-V: print version
-k password: Decryption password for encrypted volumes
```

Step 6: Recover a File (icat)

- Use the icat (inode cat) command to extract and recover a file using its inode number.
- Use inode 9871 and redirect the output to save it as a .bmp file.

```
C:\Users\sleuthkit-4.14.0-win32\bin> icat.exe -o 63 "C:\Forensics_Lab\4Dell Latitude CPi.E01" 9871 > C:\Forensics_Lab\RECOVERED_Mr_Evil.bmp
```

```
C:\Users\sleuthkit-4.14.0-win32\bin> icat.exe -o 63 "C:\Forensics_Lab\4Dell Latitude CPi.E01" [inode_number] > C:\Forensics_Lab\RECOVERED_FILE.txt
Invalid inode address: [inode_number]
usage: icat.exe [-hRsvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] image [images] inum[-typ[-id]]
-h: Do not display holes in sparse files
-r: Recover deleted file
-R: Recover deleted file and suppress recovery errors
-s: Display slack space at end of file
-i imgtype: The format of the image file (use '-i list' for supported types)
-b dev_sector_size: The size (in bytes) of the device sectors
-f fstype: File system type (use '-f list' for supported types)
-o imgoffset: The offset of the file system in the image (in sectors)
-P pooltype: Pool container type (use '-P list' for supported types)
-B pool_volume_block: Starting block (for pool volumes only)
-S snap_id: Snapshot ID (for APFS only)
-v: verbose to stderr
-V: Print version
-k password: Decryption password for encrypted volumes
```

Step 7: Create an Event Timeline (mactime)

- Generate a timeline of all file activity on the system.

(a) Generate a Body File

This file records MAC (Modified, Accessed, Changed) times for all files.

(b) Generate the Timeline Report

Use the mactime.pl script to convert the body file into a readable chronological report.

```
C:\Users\sleuthkit-4.14.0-win32\bin>mactime.pl -b C:\Forensics_Lab\body.txt > C:\Forensics_Lab\timeline.txt  
C:\Users\sleuthkit-4.14.0-win32\bin>  
[main 2025-10-24T16:43:10.884Z] updateURL https://api2.cursor.sh/updates/api/update/win32-x64/cursor/1.7.44/c3c1f9f91fe32f3f385cf6a85146cd829097bfd4f971d12db3cdfc35742429b4/stable  
[main 2025-10-24T16:43:10.885Z] update#setState idle  
C:\Users\sleuthkit-4.14.0-win32\bin>[main 2025-10-24T16:43:29.317Z] Extension host with pid 18920 exited with code: 0, signal: unknown.
```

Result:

By following the above steps, The Sleuth Kit (TSK) was successfully used to analyze the disk image 4Dell Latitude CPI.E01. The following outcomes were achieved:

- The partition table was listed using the mmls command, and the correct partition offset (63) was identified.
- File system metadata was examined using fsstat.
- All files and directories, including deleted ones, were listed using fls.
- Metadata of a specific file (inode 9871) was viewed using istat.
- The file Mr. Evil.bmp was recovered using icat.
- A complete chronological timeline of system activity was generated using mactime.

All evidence files and recovered data were saved in the C:\Forensics_Lab folder for further documentation and reporting.

Conclusion:

This experiment demonstrates the practical use of The Sleuth Kit (TSK) for digital forensic analysis. By using its command-line tools, investigators can:

- Examine disk partitions and file systems,
- Recover deleted or hidden files, and
- Reconstruct timelines of file system activity.

Therefore, TSK serves as a dependable open-source toolkit for forensic disk image examination and evidence recovery.